

Bringing AI To Edge: From Deep Learning's Perspective

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Abstract—Edge computing and artificial intelligence (AI), especially deep learning for nowadays, are gradually intersecting to build a novel system, called *edge intelligence*. However, the development of edge intelligence systems encounters some challenges, and one of these challenges is the *computational gap* between computation-intensive deep learning algorithms and less-capable edge systems. Due to the computational gap, many edge intelligence systems cannot meet the expected performance requirements. To bridge the gap, a plethora of deep learning techniques and optimization methods are proposed in the past years: light-weight deep learning models, network compression, and efficient neural architecture search. Although some reviews or surveys have partially covered this large body of literature, we lack a systematic and comprehensive review to discuss all aspects of these deep learning techniques which are critical for edge intelligence implementation. As various and diverse methods which are applicable to edge systems are proposed intensively, a holistic review would enable edge computing engineers and community to know the state-of-the-art deep learning techniques which are instrumental for edge intelligence and to facilitate the development of edge intelligence systems. This paper surveys the representative and latest deep learning techniques that are useful for edge intelligence systems, including hand-crafted models, model compression, hardware-aware neural architecture search and adaptive deep learning models. Finally, based on observations and simple experiments we conducted, we discuss some future directions.

Index Terms—Deep learning, model optimization, edge computing, neural architecture search

I. INTRODUCTION

In 2012, AlexNet [1] broke the record of ImageNet LSVRC contest [2], improving the prediction accuracy by a large margin of 10%, which marked the milestone for the booming development of deep learning (DL) techniques, or more specifically deep neural network (DNN) [3]. In the subsequent years, as the increasing number of DNN applications [4] emerges, designing DNN-based systems attracts a lot of attention and

efforts from academia and industry, spanning from advanced image manipulation and enhancement and convenient voice assistant on mobile phones to Level-4 autonomous driving systems on automobiles [5]. AlphaGo [6] is another prominent example which defeated the top professional player in the ancient Chinese game GO, deemed impossible before due to its extremely high complexity.

The successful application of DNN models relies on two stages: **training** and **inference/deployment**. Training indicates the procedure of learning a predictive model from a huge amount of labeled data, while inference denotes the procedure of using the trained model upon new data to make an accurate prediction. The training procedure involves a complex learning process requiring powerful computational units and a huge amount of data and spending considerable time. For instance, training ResNet50 [7] with ImageNet on 8 Nvidia Tesla P100 takes 29 hours [8], where ImageNet dataset [2] has 1.28 M images of 1000 classes for training. Recently, researchers also measure the environmental effect of the training procedure and indicate that training DNN consumes excessive energy and emits substantial CO₂ [9] and environment-sustainable DNN models should be considered.

The trained DNN model is exploited to make prediction on new data, and this procedure is usually called DNN *inference*. Many companies implement their DNN inference on servers to provide convenient services for their customers such as voice-assistant, machine translation, image retrieval, etc [10]. However, inference from cloud suffers from *responsiveness* (e.g., latency) and *privacy* issues which are critical in some scenarios. For instance, DNNs are seen to be a pivotal technique for self-driving cars, in which most of executions are subject to highly rigorous real-time constraints [5], [11], but the high and uncertain communication overhead makes it difficult to satisfy the temporal requirement. In addition, some inference is conducted on confidential data, e.g., manufacturing and production data, and uploading these data to cloud may risk data leakage which will severely hurt their business. To address these concerns, *edge computing* is proposed [12].

Edge computing systems are placed at the proximity of data producer, like sensors, end-users, etc, to rapidly and locally process data. The emergence of edge computing paves the way for pervasive intelligent systems [13]. IEEE Computer Society identifies ‘Artificial Intelligence at the edge’ as one of the top 12 technology trends to reach adoption in 2020 [14]. Edge systems have a broad spectrum of hardware features, from powerful computation units in communication stations and self-driving cars to small and battery-supplied smart phones and wearable devices. Fig. 1 plots the overview of edge

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